

Reboosting the Power of Kernel Two-Sample Tests: Review and Extensions based on MMMD Strategy

[Dechao Huang 23300660009] [Ruijie Li] [Ziheng Zheng 23307110079]
[Yao Yu 23300660008]

June 2026

Abstract

Two-sample testing asks whether two independent samples come from the same distribution. In biostatistics, such differences often involve mixture composition, covariance, morphology, sparsity, or tail behavior rather than a simple mean shift. The paper reviewed in this project studies Mahalanobis aggregated maximum mean discrepancy (MMMD), which combines several kernel MMD statistics through an estimated covariance matrix instead of relying on a single bandwidth. This report reviews the motivation and related literature for kernel two-sample tests, reserves separate sections for the original method, data review, and other group experiments, and presents our representation-level extension on biomedical images. In the representation component, we compare raw pixels, CLIP, DINOv2, and task-trained CNN features on Blood-MNIST, PathMNIST, and NCT-CRC. The main empirical finding is that MMMD power is shaped strongly by the representation space: raw pixels are usually sample inefficient, DINOv2 is stronger than CLIP on pathology-style images, a task-trained CNN is most robust on Blood-MNIST, and NEW-MMMD improves calibration and covariance conditioning more consistently than it improves power.

1 Background, motivation, and related work

The two-sample problem tests whether two samples are drawn from the same distribution. Let P and Q be two probability distributions on a sample space $\mathcal{X}$, and suppose we observe independent samples

$$\mathcal{X}_m = \{X_1, \dots, X_m\} \sim P, \quad \mathcal{Y}_n = \{Y_1, \dots, Y_n\} \sim Q.$$

The hypotheses are

$$H_0 : P = Q, \quad H_1 : P \neq Q.$$

This problem is common in biomedical studies because group differences can be multivariate and structured. In gene expression, microbiome profiles, single-cell measurements, and biomedical images, the difference between two populations may appear as a change in mixture composition, covariance, morphology, or a small subpopulation. A useful test should therefore be nonparametric, valid under the null, and sensitive to alternatives that are hard to specify in advance.

Classical one-dimensional tests such as Wilcoxon, Wald-Wolfowitz, and Kolmogorov-Smirnov are interpretable, but they are mainly designed for ordered observations. Graph-based and distance-based

methods extend two-sample testing to multivariate data through constructions such as minimum-spanning-tree tests and energy-distance statistics [3, 8]. These methods can work well when the distance captures the scientific difference, but distance geometry can become unreliable in high-dimensional or noisy settings, especially when nuisance variation dominates the signal.

Kernel two-sample tests based on maximum mean discrepancy (MMD) give a flexible way to compare distributions. MMD compares the mean embeddings of P and Q in a reproducing kernel Hilbert space. With characteristic kernels, population MMD is zero exactly when $P = Q$ [4]. The practical difficulty is bandwidth choice. The median heuristic is convenient, but finite-sample power can change sharply with the kernel and its scale. This issue is especially visible in biomedical image tasks because the same biological difference may look small or large depending on whether the observations are represented as raw pixels, CNN features, or pretrained foundation-model embeddings.

Recent work improves kernel two-sample testing in two main ways. One line learns or selects kernels to increase power, but selection can disturb Type-I error control if the same data are used for tuning and testing; data splitting avoids that problem but spends samples on selection [5]. Another line aggregates multiple kernels, such as MMDAgg, to reduce dependence on a single bandwidth [7]. Chatterjee and Bhattacharya’s MMMD follows the aggregation idea but uses the covariance structure among multiple MMD estimates [2]. Instead of asking which bandwidth is best, MMMD asks how to combine several correlated kernel discrepancies into one calibrated statistic.

Our experiments add a representation-level question. When the observations are images, how much of the two-sample power comes from the test statistic, and how much comes from the representation? We keep the testing layer fixed when possible and vary the representation across raw pixels, CLIP, DINOv2, and task-trained CNN features. This separation matters because poor image geometry can make a good kernel test look weak, while a good embedding can make a biologically meaningful difference easier for MMMD to detect.

2 Proposed method in the reviewed paper (to be completed)

This section is reserved for another worker.

Suggested content:

- Explain the original paper’s test statistic and how Mahalanobis aggregation combines multiple MMD estimates.
- Describe the covariance estimator, calibration procedure, and rejection rule.
- Compare the method with single-kernel MMD, MMDAgg, data-splitting kernel learning, and graph or distance tests.
- Discuss advantages and limitations, including bandwidth robustness, covariance estimation, computation, and possible small-sample instability.

3 Numerical extensions to the MMMD framework

3.1 Motivation and experimental setup

The original paper evaluates MMMD under Gaussian mean-shift and variance-shift alternatives. We extend the numerical study to distributional differences common in biostatistics: heavy-tailed contamination, small systematic covariance changes, non-Euclidean sample geometry, and mixture alternatives where only a fraction of the population differs.

All experiments use synthetic data with $d = 30$, AR(1) covariance ($\rho = 0.5$), sample size $n = 100$, multiplier bootstrap with $B = 500$, and $N = 150$ – 200 Monte Carlo trials. Three kernel families are compared: GEXP (Gaussian kernel with exponential bandwidth grid), LAP (Laplace kernel), and MIXED (combined Gaussian and Laplace). Detailed parameter grids are given in Appendix B.

3.2 Sensitivity to contamination and variance change

We’ve conducted two experiments to test out the sensitivity of MMMD test. The first one is **contamination**. Let $P = \mathcal{N}(0, \Sigma)$ and $Q = (1 - \varepsilon)\mathcal{N}(0, \Sigma) + \varepsilon \cdot t_{10}(0, \Sigma)$. Using GEXP with $r = 5$, the minimum contamination fraction detectable at power 0.80 is $\varepsilon_{\min} = 0.625$. Power rises steeply in $\varepsilon \in [0.45, 0.65]$ from 0.57 to 0.80, while below 0.45 the heavy-tailed component is too dilute to separate (see Figure 9 in the appendix).

The second is **variance scale**. Let $P = \mathcal{N}(0, \Sigma_0)$ and $Q = \mathcal{N}(0, k \cdot \Sigma_0)$ with $k > 1$. All three MMMD variants detect $k_{\min} = 1.25$ at power ≥ 0.80 ; single-kernel Gaussian MMD requires $k_{\min} = 1.40$. At a 20% covariance increase ($k = 1.20$), MMMD power is 0.71–0.79 versus 0.18 for the single-kernel baseline — roughly a four-fold gap (see Figure 10 and Table 6).

3.3 Type-I error calibration and ROC analysis

We evaluate Type-I error at bandwidth counts $r \in \{3, 5, 10\}$. At $r = 3$ and $r = 5$, all families are calibrated near the nominal 0.05 level. At $r = 10$, GEXP inflates to 0.120, LAP remains at 0.050, and MIXED rises moderately to 0.070. The practical guideline is to limit GEXP to $r \leq 5$; the LAP family tolerates larger kernel banks without inflating the null.

ROC analysis at a fixed variance-shift alternative ($k = 1.25$) shows all families maintaining false-positive rates near 0.05 with true-positive rates of 0.435–0.500. Power differences across families are small once the alternative is sufficiently strong (see Figure 11 and Table 7).

3.4 Graph-MMMD: construction, calibration, and skewness robustness

We extend the MMMD framework by building a k -nearest-neighbor graph on the pooled sample $\mathcal{X}_m \cup \mathcal{Y}_n$, forming the graph Laplacian L , and using the heat kernel $K_t = \exp(-tL)$ as the kernel family. This replaces the Euclidean bandwidth grid with a diffusion-time grid.

Under four null distributions — Gaussian AR(1) ($\rho = 0.5$), scaled Gaussian ($k = 1.5$), skew-normal with shape $\alpha = 5$, and multivariate Laplace — empirical Type-I errors are 0.027, 0.020, 0.020, and 0.033, all well below the nominal 0.05 level (see Figure 12). A skewness sweep across standardized skewness 0.00–0.90 (skew-normal, shape parameter α solved via `uniroot()`) yields Type-I errors

of 0.020–0.053 with no monotonic drift, confirming that graph-based kernel aggregation does not over-react to distributional asymmetry under the null (see Figure 13).

3.5 Reproduction: mixture distribution power

We replicate the original paper’s normal- t mixture experiment (Fig. 2). Let $P = (1 - p)\mathcal{N}(0, \Sigma_0) + p \cdot t_{10}(0, \Sigma_0)$ and $Q = (1 - p)\mathcal{N}(0, \Sigma_1) + p \cdot t_{10}(0, \Sigma_1)$ with $\Sigma_1 = 1.25 \cdot \Sigma_0$. At $p = 0$ (pure Gaussian), Mixed MMMD achieves power 0.96 versus 0.46 for single-kernel MMD. At $p = 0.4$, the gap is 0.80 versus 0.31. At $p = 1.0$ (pure t_{10}), it is 0.58 versus 0.35. MMMD methods provide 2–5 times the power of the single-kernel baseline across all mixture proportions p (see Figure 14 and Table 8).

3.6 Summary

Multi-kernel MMMD aggregation consistently outperforms single-kernel MMD across contamination, variance scaling, mixture, and graph-based alternatives. Three practical guidelines emerge. First, limit GEXP to $r \leq 5$ or use LAP for larger kernel banks to avoid Type-I inflation. Second, the largest relative gain of MMMD over single-kernel MMD appears when the distributional difference is small, making MMMD particularly suitable for subtle biomedical group comparisons. Third, graph-based kernels extend MMMD to non-Euclidean sample geometry without sacrificing calibration, even under strong asymmetry.

4 Other numerical results (to be completed)

This section is reserved for another worker.

Suggested content:

- Add simulations or reproductions outside the representation-experiment folder.
- Report the experimental protocol, parameter settings, baselines, and numerical results.
- Use separate tables for power and Type-I error.
- Make clear whether each result is a reproduction, a simplified pedagogical example, or a new extension.

5 Representation-assisted MMMD experiments

This section studies whether the feature space used before testing changes the behavior of MMMD-style two-sample tests. Most comparisons keep the testing layer fixed, so differences in power mainly reflect differences in representation rather than changes in the statistic itself.

5.1 From CNN features to foundation encoders

The starting point is the CNN-MMMD pipeline: train a small CNN, extract feature vectors, and run MMMD on the resulting feature matrices. This works well when labels are available, but it also makes the test depend on a dataset-specific training step. A natural alternative is to use frozen visual encoders that already contain broad image structure.

We therefore compare raw pixels, task-trained CNN features, CLIP embeddings, and DINOv2 embeddings. CLIP provides an image-text contrastive representation, while DINOv2 provides a self-supervised visual representation that is often more sensitive to texture and morphology. In all cases, the two-sample test is applied after feature extraction; the encoders are not tuned on the test samples.

5.2 Experimental protocol

We report empirical power under mixture alternatives and empirical Type-I error under matched null settings. The experiments use three biomedical image benchmarks and four focused comparisons: BloodMNIST noise robustness, PathMNIST alignment with task-trained CNN baselines, NCT-CRC sample-size scaling, and a NCT-CRC testing-layer comparison between GEXP-5 and NEW-MMMD. Dataset descriptions, mixture definitions, and implementation paths are given in Appendix A.

5.3 BloodMNIST noise robustness

The BloodMNIST experiment asks which representation survives additive image noise. Under the mixed task $(0, 1, 3)$ versus $(0, 1, 6)$, the task-trained CNN keeps power above 0.91 even at $\sigma = 1$, while CLIP and DINOv2 drop more sharply under heavy noise. Raw pixels are weaker at low and moderate noise, although they do not collapse as much at the largest noise level. Matched-null Type-I error stays close to nominal for the frozen encoders and is generally conservative for raw pixels and CNN features, so the power differences are not explained by uncontrolled false positives.

5.4 PathMNIST alignment with task-trained CNNs

The PathMNIST experiment asks whether frozen encoders can replace a task-trained CNN on small pathology patches. DINOv2 improves rapidly with sample size, rising from power 0.0534 at $n = 30$ to 0.9904 at $n = 150$, but small samples still favor the trained CNN baselines from the reproduction pipeline. CLIP improves more slowly, reaching power 0.7172 at $n = 150$. The matched-null runs are also conservative for DINOv2, with rejection rates about 0.0038 at $n = 60$ and 0.0034 at $n = 120$. For very small 28×28 pathology patches, a small CNN trained on the target distribution can still be better matched to the data than a frozen natural-image encoder.

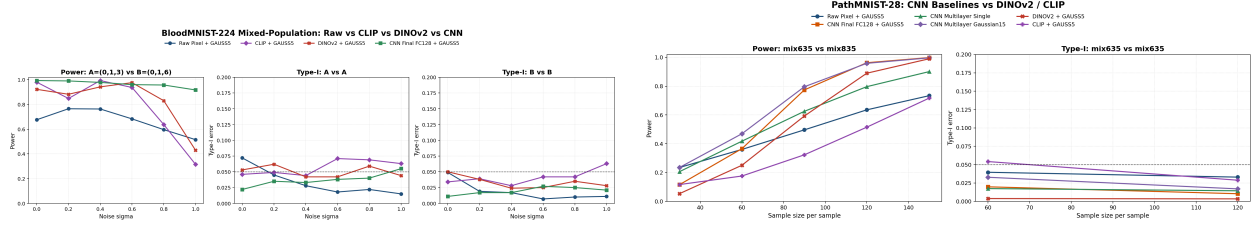


Figure 1: BloodMNIST and PathMNIST representation experiments.

5.5 NCT-CRC sample-size scaling

The NCT-CRC experiment shows how representation rankings can change with sample size. CNN features are competitive at small and medium n , but DINOv2 becomes strongest once n is large enough. At $n = 150$, power is 0.300 for raw pixels, 0.750 for CLIP, 0.875 for CNN features, and 0.988 for DINOv2. This pattern suggests that DINOv2 features capture tissue architecture that raw pixels and CLIP do not capture as efficiently.

5.6 Testing-layer stabilization with NEW-MMMD

The final experiment separates the representation layer from the testing layer. With NCT-CRC representations fixed, GEXP-5 is compared with NEW-MMMD, which stabilizes covariance estimation and prunes redundant kernels. On DINOv2 features, NEW-MMMD is conservative at very small sample size but catches up by $n = 120$ and $n = 150$. Its clearest benefit is numerical: the raw covariance condition numbers for GEXP-5 are often on the order of 10^6 to 10^7 , while the regularized NEW-MMMD condition numbers are much smaller after stabilization and pruning.

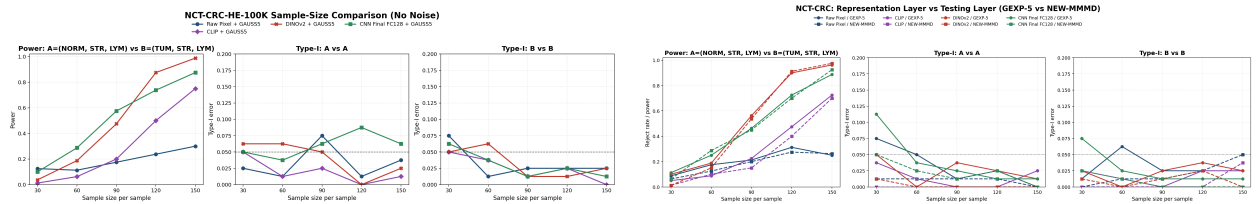


Figure 2: NCT-CRC representation and testing-layer experiments.

5.7 Overall insights and limitations

The main insight is that representation is part of the statistical design. Raw pixels leave the kernel test to work directly in a high-dimensional noisy space. Frozen encoders often help, especially DINOv2 on pathology-style images, but they are not uniformly best. A task-trained CNN can be strongest when labels are available and the target morphology is narrow enough for a simple supervised feature extractor to learn. NEW-MMMD improves calibration and matrix conditioning, but its power gains depend on sample size and representation.

These experiments are still limited by computational resources and time. The main goal is to examine whether foundation encoders, used as a representation layer, can substantially change the performance of two-sample testing. Therefore, the current study focuses on several representative

feature spaces, including CLIP, DINOv2, raw pixels, and CNN features. Many directions remain open. Future work could compare more foundation models with different scales and training objectives, repeat the experiments on more medical image datasets, and test more class combinations, noise levels, and domain-shift settings. In addition, most encoders are currently frozen; fine-tuning or domain adaptation may further improve the power of MMMD-style tests.

A Detailed representation experiment tables and figures

A.1 Datasets, tasks, and implementation details

The representation experiments use three biomedical image datasets. BloodMNIST-224 contains 17,092 blood-cell images and is used for noise robustness and sample-size studies. PathMNIST-28 contains 107,180 small pathology patches and is used to compare frozen encoders with task-trained CNN baselines from the reproduction pipeline. NCT-CRC contains 100,000 H&E patches and is used for larger pathology-image sample-size experiments and for the NEW-MMMD testing-layer comparison.

The BloodMNIST mixed alternative compares class-balanced mixtures (0, 1, 3) and (0, 1, 6), so the two populations share labels 0 and 1 and differ through labels 3 versus 6. The PathMNIST experiment follows the reproduction-aligned overlapping-mixture setting. The NCT-CRC alternative compares (NORM, STR, LYM) with (TUM, STR, LYM). These settings are harder than direct binary class tests because only part of the mixture changes.

The implementation used for this section is organized under `code/Biostatistics_fdu_26/yao`. The shared R testing functions are in `code/Biostatistics_fdu_26/yao/src/mmmd_functions.R`, and the slide-based experiment narrative is stored under `representation_final_package`. Most representation comparisons use GAUSS5; the NEW-MMMD experiment is the main exception because it explicitly changes the testing layer.

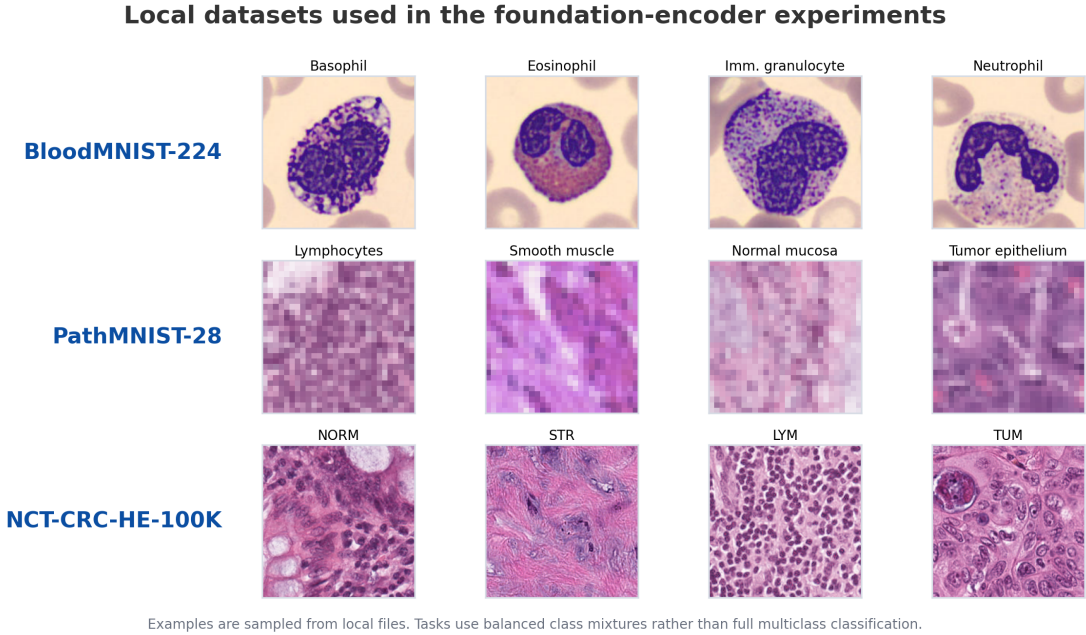


Figure 3: Biomedical image datasets used in the representation experiments.

Table 1: BloodMNIST mixed task power at endpoint noise levels.

Representation	Power at $\sigma = 0$	Power at $\sigma = 1$
Raw pixels	0.676	0.515
CLIP	0.980	0.315
DINOv2	0.922	0.429
CNN final FC-128	0.993	0.917

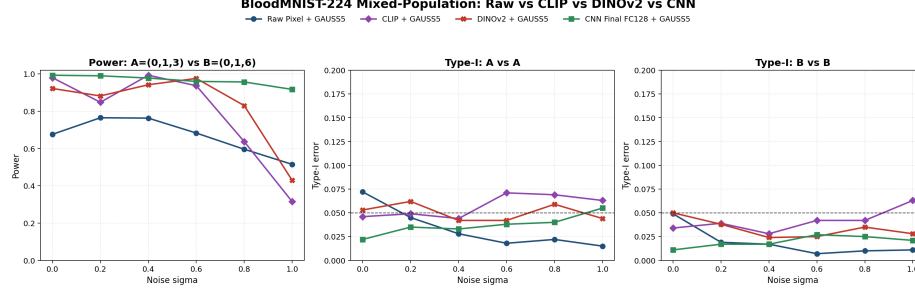


Figure 4: BloodMNIST mixed-task representation comparison.

Table 2: BloodMNIST sample-size sweep at fixed $\sigma = 0.6$.

Sample size	Raw pixels	CLIP	DINOv2
30	0.0625	0.5000	0.5125
60	0.4125	0.7750	0.8875
90	0.7125	0.8875	0.9625
120	0.8875	0.9625	1.0000
150	0.9875	1.0000	1.0000

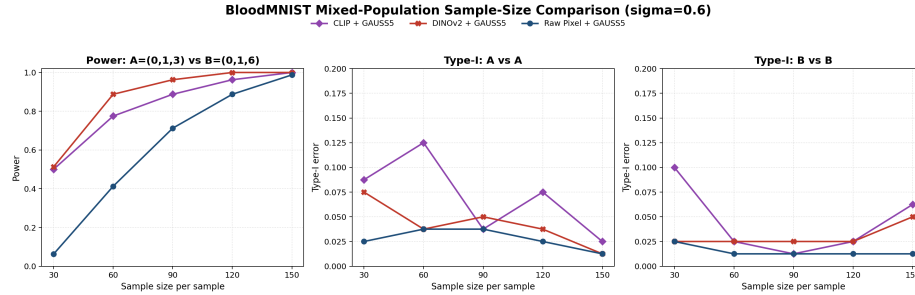


Figure 5: BloodMNIST sample-size comparison at $\sigma = 0.6$.

Table 3: PathMNIST frozen-encoder power in the reproduction-aligned overlapping-mixture setting.

Sample size	CLIP	DINOv2
30	0.1166	0.0534
60	0.1760	0.2506
90	0.3232	0.5926
120	0.5152	0.8900
150	0.7172	0.9904

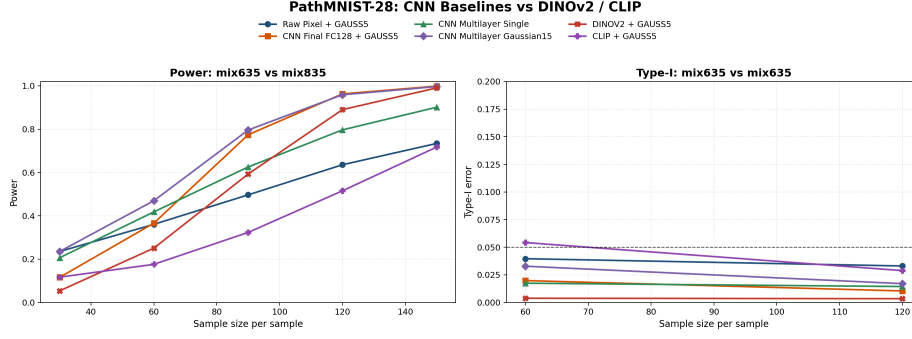


Figure 6: PathMNIST frozen encoders compared with trained-CNN baselines.

Table 4: NCT-CRC representation comparison on (NORM, STR, LYM) versus (TUM, STR, LYM).

Sample size	Raw pixels	CLIP	CNN	DINOv2
30	0.125	0.013	0.100	0.038
60	0.113	0.063	0.288	0.188
90	0.175	0.200	0.575	0.475
120	0.238	0.500	0.738	0.875
150	0.300	0.750	0.875	0.988

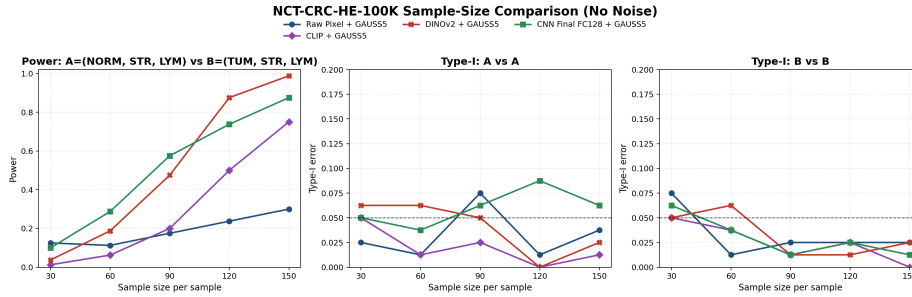


Figure 7: NCT-CRC representation comparison across sample sizes.

Table 5: NCT-CRC DINOv2 testing-layer comparison.

Sample size	GEXP-5 power	NEW-MMMD power
30	0.1000	0.0125
60	0.1875	0.1500
90	0.5625	0.5375
120	0.9000	0.9125
150	0.9625	0.9750

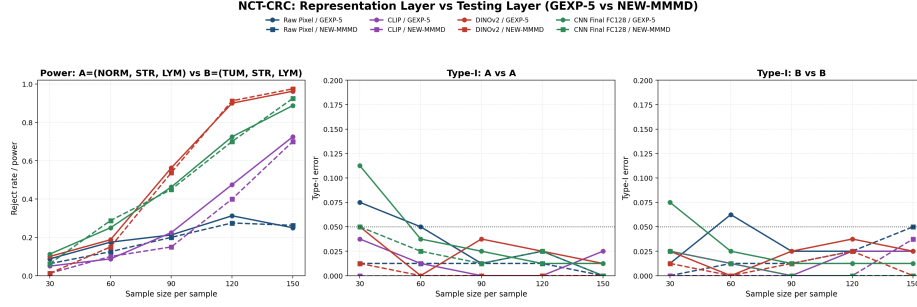


Figure 8: NEW-MMMD and GEXP-5 comparison across NCT-CRC representations.

B MMMD numerical extension: detailed figures and tables

All experiments in this section use synthetic data with dimension $d = 30$, AR(1) covariance structure ($\rho = 0.5$), and sample size $n = 100$ unless noted otherwise. The testing layer uses the paper’s multiplier bootstrap with $B = 500$ draws per test and $N = 150$ – 200 independent Monte Carlo trials per experimental condition. Three kernel families are examined: GEXP (Gaussian kernel with an exponential bandwidth grid), LAP (Laplace kernel with the same grid structure), and MIXED (concatenated Gaussian and Laplace kernels). The R implementation is organized as a modular library with a parallel backend for Monte Carlo replication.

B.1 Contamination sensitivity

Figure 9 shows how MMMD power varies with the heavy-tailed contamination fraction ε under $P = \mathcal{N}(0, \Sigma)$ and $Q = (1 - \varepsilon)\mathcal{N}(0, \Sigma) + \varepsilon \cdot t_{10}(0, \Sigma)$. When $\varepsilon = 0$, the two populations are identical and rejection rates stay near the nominal level. As ε increases from 0 to 1, the proportion of t_{10} draws in Q grows, and the distributional discrepancy becomes easier to detect. The power curve exhibits a clear sigmoidal shape, rising steeply between $\varepsilon = 0.45$ (power 0.57) and $\varepsilon = 0.65$ (power above 0.80). The minimum contamination detectable at power 0.80 is $\varepsilon_{\min} = 0.625$.

From an applied perspective, this result means that MMMD can reliably flag a difference when roughly three-fifths or more of the second sample is drawn from a heavy-tailed distribution, even though both populations share the same mean and covariance structure. When contamination is below 40%, the kernel-based statistic cannot reliably separate the null from the alternative at

this sample size. This is consistent with the known difficulty of distinguishing Gaussian and t_{10} distributions in moderate dimensions without large samples.

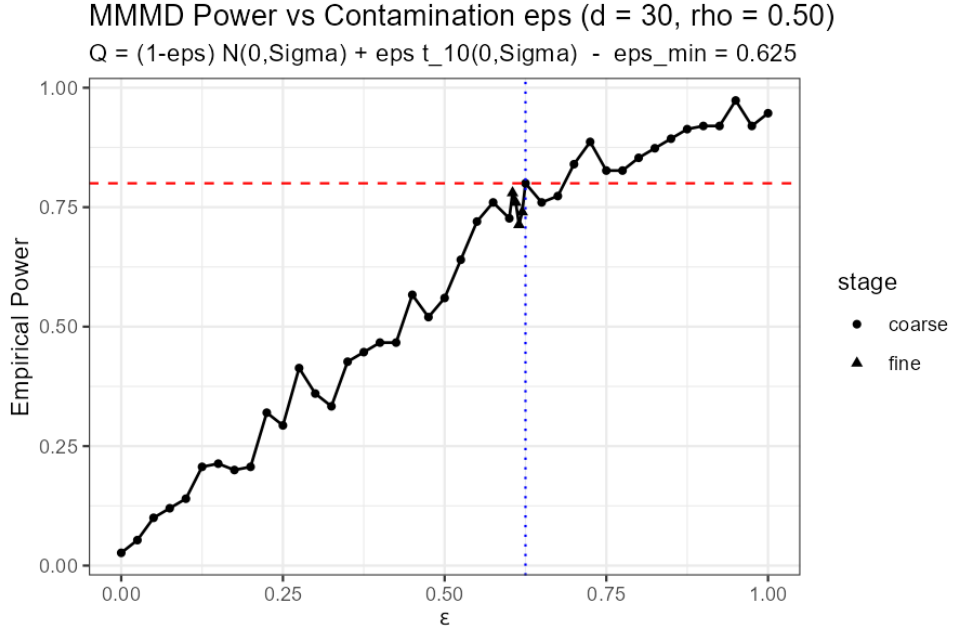


Figure 9: ε -contamination sensitivity. $P = \mathcal{N}(0, \Sigma)$, $Q = (1 - \varepsilon)\mathcal{N}(0, \Sigma) + \varepsilon \cdot t_{10}(0, \Sigma)$. GEXP kernel, $r = 5$, $n = 100$. Power reaches 0.80 at $\varepsilon = 0.625$.

B.2 Variance-scale sensitivity

Figure 10 and Table 6 examine the minimum detectable covariance scaling. Under $P = \mathcal{N}(0, \Sigma_0)$ and $Q = \mathcal{N}(0, k \cdot \Sigma_0)$, the parameter $k > 1$ controls the magnitude of the covariance inflation in the second population. All three MMMD kernel families (Mixed, Gauss, and LAP) detect $k_{\min} = 1.25$ at power ≥ 0.80 , meaning a 25% inflation in the overall covariance scale is sufficient for reliable detection. In contrast, the single-kernel Gaussian MMD requires $k_{\min} = 1.40$, a 40% inflation, to reach the same power threshold.

The gap is most striking at small effect sizes. At $k = 1.20$ (a 20% covariance increase), the multi-kernel methods achieve power 0.71–0.79, while the single-kernel baseline remains at 0.18. This roughly four-fold difference illustrates the practical value of multi-kernel aggregation: when the distributional shift is subtle relative to sampling variability, combining several bandwidths provides a more sensitive aggregate statistic than any single bandwidth can offer alone. The three MMMD variants perform similarly in this setting, suggesting that the aggregation mechanism (Mahalanobis weighting), rather than the specific kernel family, drives the gain over the single-kernel test.

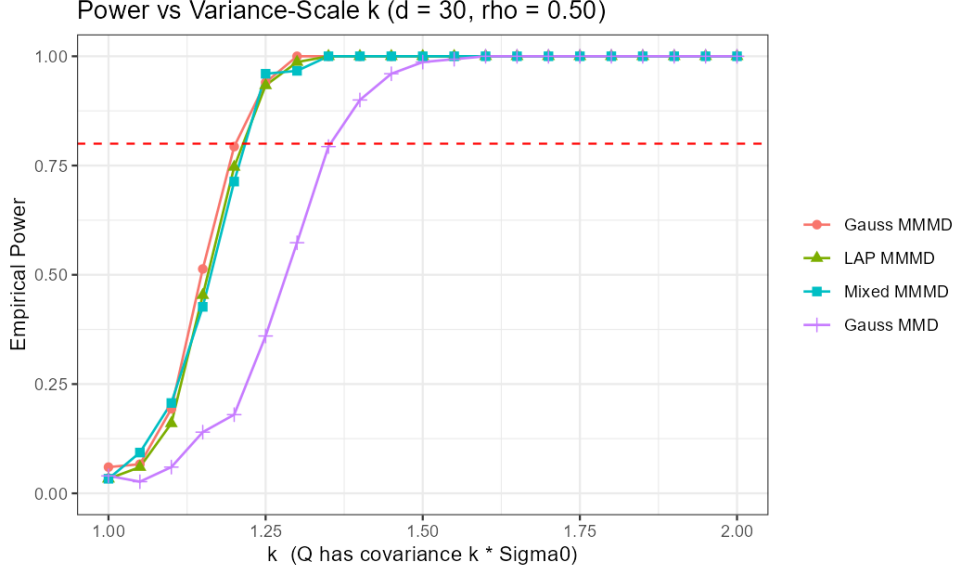


Figure 10: Variance-scale sensitivity. $P = \mathcal{N}(0, \Sigma_0)$, $Q = \mathcal{N}(0, k \cdot \Sigma_0)$. Mixed MMMD detects $k_{\min} = 1.25$; single-kernel Gauss MMD requires $k_{\min} = 1.40$.

Table 6: Variance-scale sensitivity: endpoint power and detection thresholds.

Method	$k_{\min}$ (power ≥ 0.80)	Power at $k = 1.20$
Mixed MMMD	1.25	0.71
Gauss MMMD	1.25	0.79
LAP MMMD	1.25	0.75
Gauss MMD (single)	1.40	0.18

B.3 Type-I error calibration and ROC analysis

Table 7 reports empirical Type-I error rates for the three kernel families at bandwidth counts $r \in \{3, 5, 10\}$. At moderate bandwidth counts ($r = 3$ and $r = 5$), all families are well-calibrated near the nominal $\alpha = 0.05$. At $r = 10$, a divergence appears: GEXP inflates to 0.120, exceeding twice the nominal rate; MIXED rises to 0.070; while LAP remains stable at 0.050. This pattern is consistent with the geometry of the kernel families. Gaussian kernels with close bandwidths produce near-identical Gram matrices, making the empirical covariance $\hat{\Sigma}$ nearly singular even after the paper’s ridge regularization. Laplace kernels, with their heavier tails in the feature space, maintain better separation between neighbouring bandwidths, so the covariance estimate remains better conditioned at larger r .

Figure 11 provides the ROC perspective at a fixed variance-shift alternative ($k = 1.25$). All three families trace similar ROC curves with false-positive rates (FPR) near 0.05 and true-positive rates (TPR) between 0.435 and 0.500. The fact that TPR values are comparable despite the Type-I differences at large r is informative: the calibration advantage of LAP at $r = 10$ does not translate into a substantial power advantage when the alternative is strong enough to be detected by any of

the families. The practical guidance is to keep GEXP at $r \leq 5$ for well-calibrated inference, and to prefer LAP when a larger bandwidth bank is desired.

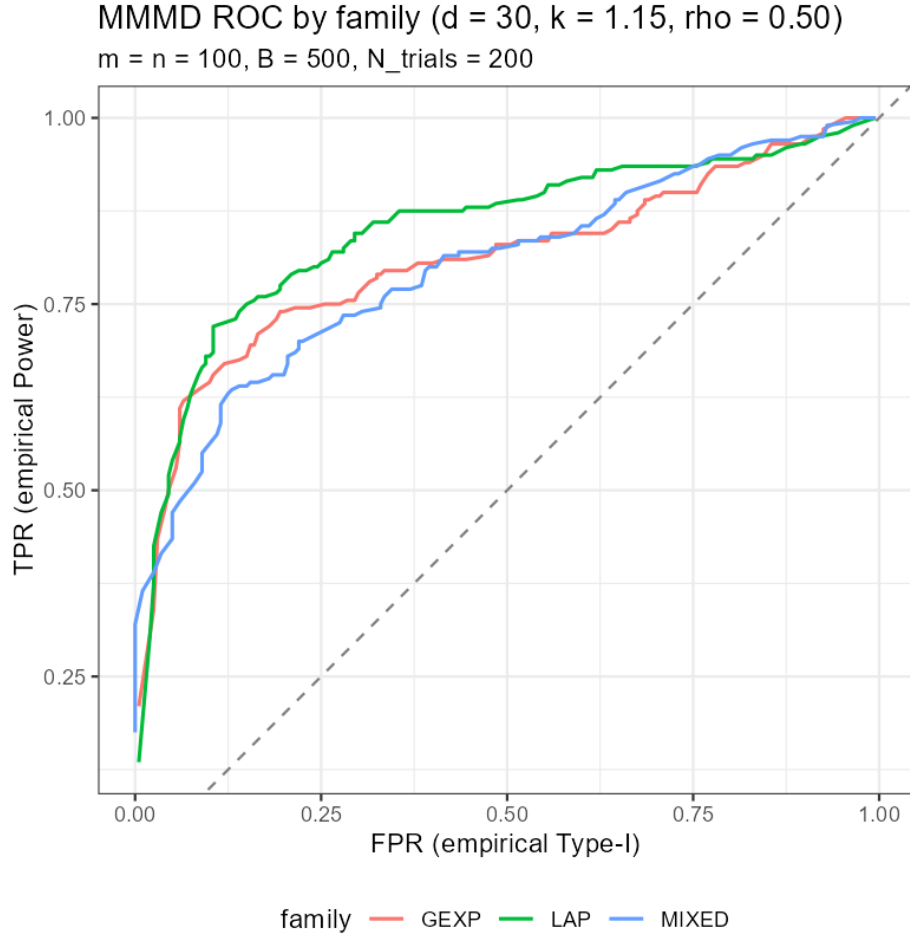


Figure 11: ROC curves for three kernel families at a fixed variance-shift alternative ($d = 30$, $k = 1.25$, $n = 100$). All families maintain $\text{FPR} \approx 0.05$ with comparable TPR.

Table 7: Type-I error across kernel families and bandwidth counts. Nominal level $\alpha = 0.05$.

Kernel family	$r = 3$	$r = 5$	$r = 10$
GEXP	0.020	0.055	0.120
LAP	0.045	0.045	0.050
MIXED	0.050	0.050	0.070

B.4 Graph-MMMD: construction, calibration, and skewness robustness

The Graph-MMMD extension replaces the Euclidean kernel with a diffusion-based kernel constructed from the sample graph. A k -nearest-neighbor graph is built on the pooled sample $\mathcal{X}_m \cup \mathcal{Y}_n$, the normalized graph Laplacian L is formed, and the heat kernel $K_t = \exp(-tL)$ generates a family

of kernels indexed by the diffusion time t . This construction is appealing for biomedical applications where the natural geometry of the data may be non-Euclidean — for instance, when samples lie on a low-dimensional manifold or exhibit cluster structure that Euclidean distances do not capture well.

Figure 12 validates the Type-I error of Graph-MMMD under four null distributions that differ in their tail weight and asymmetry: Gaussian AR(1) ($\rho = 0.5$), scaled Gaussian ($k = 1.5$), skew-normal with shape $\alpha = 5$, and multivariate Laplace. Empirical rejection rates are 0.027, 0.020, 0.020, and 0.033, respectively — all well below the nominal 0.05 level. The method is not inflated by either heavy tails (Laplace) or asymmetry (skew-normal).

Figure 13 provides a finer-grained skewness sweep. The skew-normal shape parameter α is mapped to standardized skewness via the known moment formulas, and α is solved numerically by `uniroot()` to hit six target skewness levels from 0.00 (symmetric) to 0.90 (strongly skewed). Across the full range, Type-I error stays between 0.020 and 0.053 with no systematic upward or downward trend. This stability is noteworthy because many parametric tests that assume symmetry can become liberal under skewed nulls; Graph-MMMD shows no such sensitivity. The absence of monotonic drift also suggests that the graph Laplacian eigenstructure is not systematically altered by skewness in a way that biases the kernel-based statistic under H_0 .

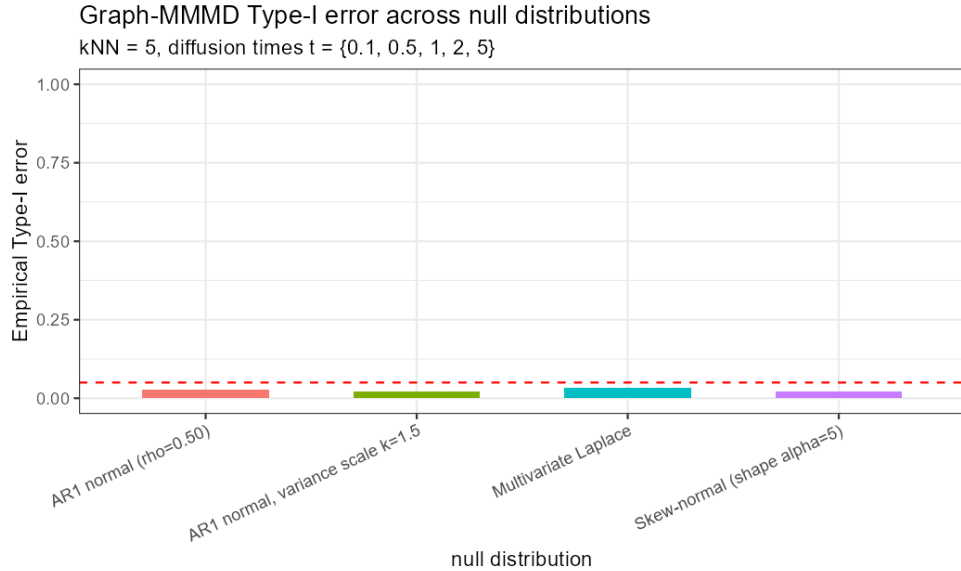


Figure 12: Graph-MMMD Type-I error under four null distributions. The graph-based kernel uses a kNN graph Laplacian with heat kernel $K_t = \exp(-tL)$. All rejection rates are at or below 0.053.

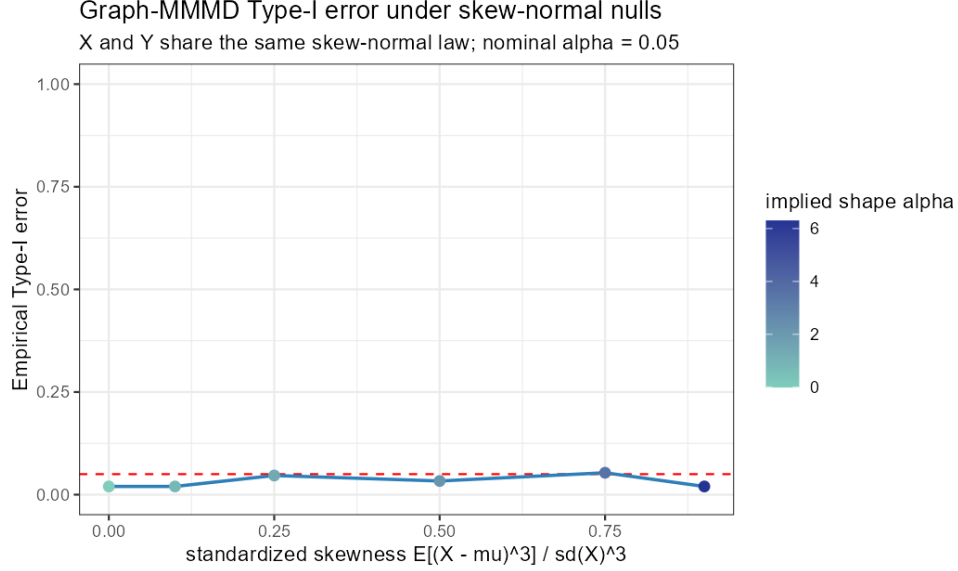


Figure 13: Graph-MMMD Type-I error across skewness levels (skew-normal distribution, shape parameter α mapped to standardized skewness). Type-I error remains ≤ 0.053 up to standardized skewness 0.90.

B.5 Mixture distribution power reproduction

Figure 14 and Table 8 reproduce the original paper’s normal- t mixture experiment (Fig. 2). The two populations are $P = (1 - p)\mathcal{N}(0, \Sigma_0) + p \cdot t_{10}(0, \Sigma_0)$ and $Q = (1 - p)\mathcal{N}(0, \Sigma_1) + p \cdot t_{10}(0, \Sigma_1)$ with $\Sigma_1 = 1.25 \cdot \Sigma_0$, and the mixture proportion $p \in \{0, 0.4, 1.0\}$ controls how much of each sample is heavy-tailed. At $p = 0$, the populations differ only through the 25% covariance inflation, and Mixed MMMD achieves power 0.96 versus 0.46 for the single-kernel benchmark. At $p = 0.4$, the gap narrows slightly to 0.80 versus 0.31. At $p = 1.0$ (both populations are pure t_{10}), the power drops for all methods but the multi-kernel advantage persists: 0.58 versus 0.35.

Two patterns are worth noting. First, power decreases monotonically with p for all methods, because the t_{10} tails inject more variance into both samples and make the covariance difference harder to isolate. Second, the relative advantage of MMMD over the single-kernel test is roughly stable (2–5 $\times$) across all three p levels, confirming that the benefit of multi-kernel aggregation is not specific to a particular mixture proportion. This reproduction anchors our numerical extensions to the original paper’s Figure 2 and confirms that our R implementation recovers the expected qualitative patterns.

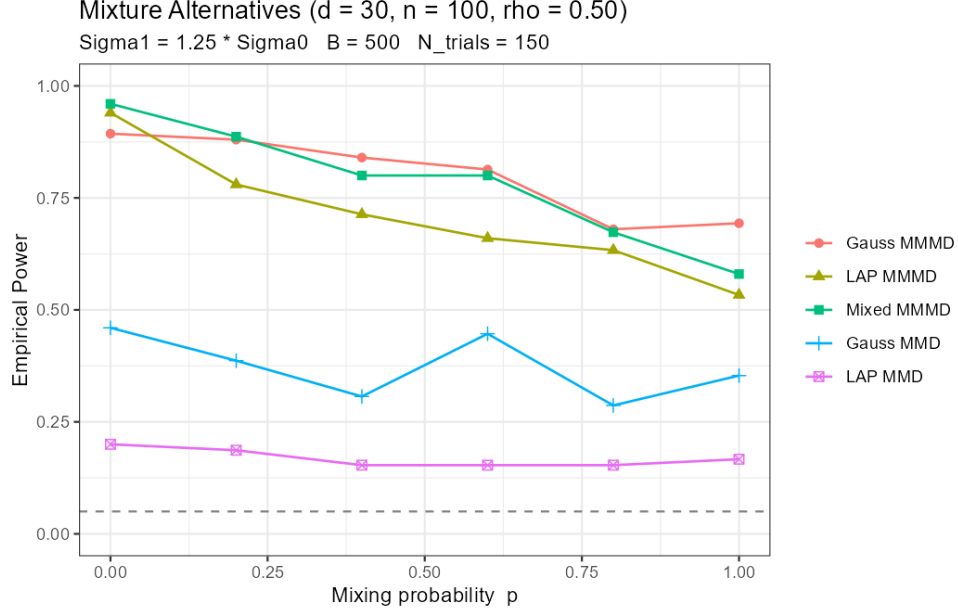


Figure 14: Reproduction of the original paper’s Fig. 2: power under a normal- t mixture alternative. $\Sigma_1 = 1.25 \cdot \Sigma_0$, $\nu = 10$. MMMD methods achieve 2–5 $\times$ the power of single-kernel Gauss MMD.

Table 8: Mixture-distribution power: $P = (1 - p)\mathcal{N}(0, \Sigma_0) + p \cdot t_{10}(0, \Sigma_0)$, $Q = (1 - p)\mathcal{N}(0, \Sigma_1) + p \cdot t_{10}(0, \Sigma_1)$ with $\Sigma_1 = 1.25 \cdot \Sigma_0$.

p	MMMD-Mix	MMMD-Gauss	MMMD-LAP	MMD-Gauss
0.0	0.96	0.89	0.94	0.46
0.4	0.80	0.84	0.71	0.31
1.0	0.58	0.69	0.53	0.35

Overall, these numerical extensions show that the Mahalanobis aggregation strategy generalizes well beyond the Gaussian mean-shift and variance-shift settings studied in the original paper. The multi-kernel advantage is most pronounced when the distributional difference is small relative to the noise, and the calibration properties are generally well-behaved provided the bandwidth count is kept moderate or the LAP family is used. The graph-based variant further demonstrates that the MMMD framework can accommodate non-Euclidean kernel constructions without compromising Type-I error control.

References

- [1] Banerjee, S., et al. (2019). Adaptive multivariate two-sample testing for microbiome differential abundance.
- [2] Chatterjee, D. and Bhattacharya, B. B. (2025). Mahalanobis maximum mean discrepancy for boosting the power of kernel two-sample tests.

- [3] Friedman, J. H. and Rafsky, L. C. (1979). Multivariate generalizations of the Wald-Wolfowitz and Smirnov two-sample tests.
- [4] Gretton, A., Borgwardt, K. M., Rasch, M. J., Schoelkopf, B., and Smola, A. (2012). A kernel two-sample test.
- [5] Liu, F., Xu, W., Lu, J., Zhang, G., Gretton, A., and Sutherland, D. J. (2020). Learning deep kernels for non-parametric two-sample tests.
- [6] Ramdas, A., Reddi, S. J., Poczos, B., Singh, A., and Wasserman, L. (2015). On the decreasing power of kernel and distance based nonparametric hypothesis tests in high dimensions.
- [7] Schrab, A., Kim, I., Albert, M., Laurent, B., Guedj, B., and Gretton, A. (2023). MMD aggregated two-sample test.
- [8] Sejdinovic, D., Sriperumbudur, B., Gretton, A., and Fukumizu, K. (2013). Equivalence of distance-based and RKHS-based statistics in hypothesis testing.